VisionForge AI – Dockerized Web Application Deployment on AWS EC2

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Section 1 - Project Overview:

VisionForge AI is a static web application designed to simulate an AI-powered image captioning platform. The website demonstrates responsive UI design using HTML, CSS, Bootstrap, and JavaScript. The application was containerized using Docker and deployed to an AWS EC2 instance using a publicly accessible HTTP server.

The objective of this project was to:

- Containerize the application using Docker
- Push the image to Docker Hub
- Deploy the container to AWS EC2
- Configure networking and security
- Verify the application is publicly accessible via EC2 public IP

Section 2 - Docker Configuration:

Screenshot #1: Project Structure and Development Environment (WebStorm)

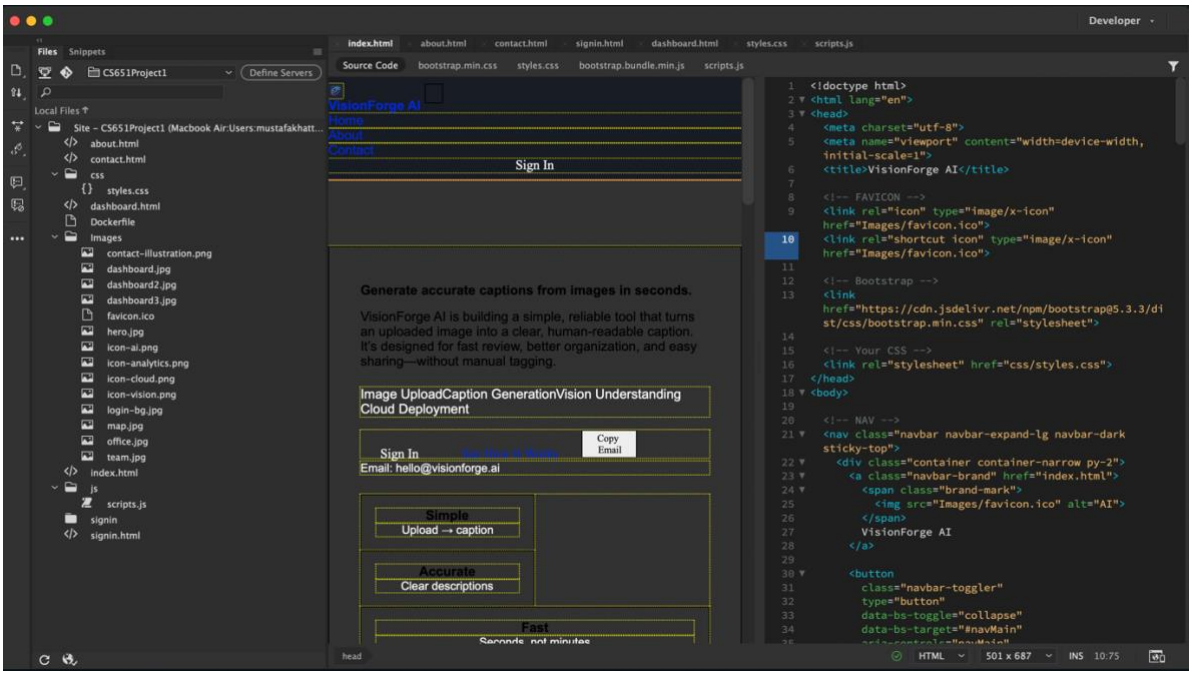

Figure 1: WebStorm development environment showing the complete VisionForge AI project structure, including HTML pages (index, about, contact, sign-in, dashboard), CSS styling files, JavaScript scripts, image assets, and Dockerfile used for containerization.

Screenshot #2: Local Docker Container Running (HTTP 200 Verification)

```
Last login: Mon Mar  2 23:17:58 on ttys001
mustafakhattak@Mustafas-MacBook-Air-4 CS651Project1 % curl -I http://localhost:8080
HTTP/1.1 200 OK
Server: nginx/1.29.5
Date: Tue, 03 Mar 2026 07:18:37 GMT
Content-Type: text/html
Content-Length: 9551
Last-Modified: Tue, 03 Mar 2026 06:06:15 GMT
Connection: keep-alive
ETag: "69a67a57-254f"
Accept-Ranges: bytes

mustafakhattak@Mustafas-MacBook-Air-4 CS651Project1 %
```

Figure 2: Verification of the locally running Docker container using curl -I http://localhost:8080. The HTTP 200 OK response confirms that the Nginx server inside the container successfully served the VisionForge AI application.

Screenshot #3: Docker Container Running Locally with Port Mapping

```
Last login: Mon Mar  2 23:17:58 on ttys001
mustafakhattak@Mustafas-MacBook-Air-4 CS651Project1 % curl -I http://localhost:8080
HTTP/1.1 200 OK
Server: nginx/1.29.5
Date: Tue, 03 Mar 2026 07:18:37 GMT
Content-Type: text/html
Content-Length: 9551
Last-Modified: Tue, 03 Mar 2026 06:06:15 GMT
Connection: keep-alive
ETag: "69a67a57-254f"
Accept-Ranges: bytes

mustafakhattak@Mustafas-MacBook-Air-4 CS651Project1 % docker ps
CONTAINER ID   IMAGE             COMMAND                  CREATED         STATUS         PORTS                                      NAMES
e260f4c9eb7b   visionforge-site  "/docker-entrypoint.…"   5 minutes ago   Up 5 minutes   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   affectionate_albattani
mustafakhattak@Mustafas-MacBook-Air-4 CS651Project1 %
```

Figure 3: Docker container running locally using the Nginx base image. The docker ps output confirms the VisionForge application is mapped from host port 8080 to container port 80 (0.0.0.0:8080 → 80), demonstrating successful container execution.

Screenshot #4: VisionForge AI Running Locally via Docker (Browser Verification)

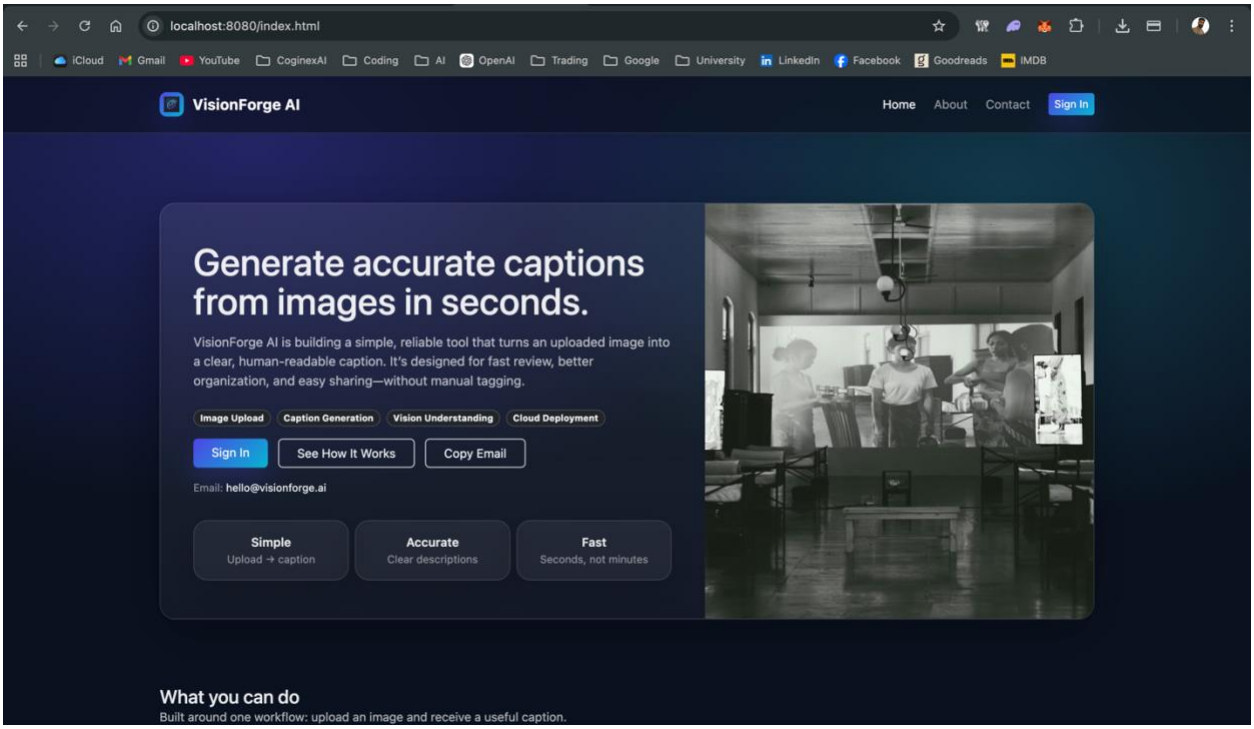

Figure 4: VisionForge AI successfully running locally through Docker at http://localhost:8080. This confirms that the containerized Nginx server properly serves all HTML, CSS, JavaScript, and image assets before cloud deployment.

Screenshot #5: Docker Hub Authentication (Login Successful)

```
Last login: Mon Mar  2 23:18:34 on ttys001
[mustafakhattak@Mustafas-MacBook-Air-4 CS651Project1 % docker ps
CONTAINER ID   IMAGE             COMMAND                  CREATED         STATUS         PORTS                                      NAMES
e260f4c9eb7b   visionforge-site  "/docker-entrypoint.…"   5 minutes ago   Up 5 minutes   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp    affectionate_albattani
[mustafakhattak@Mustafas-MacBook-Air-4 CS651Project1 % docker login
Authenticating with existing credentials... [Username: mustafakhattak]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded
mustafakhattak@Mustafas-MacBook-Air-4 CS651Project1 %
```

Figure 5: Successful authentication to Docker Hub using docker login, confirming access to push the VisionForge Docker image to the remote repository for cloud deployment.

Screenshot #6: Project Directory Verification Before Docker Build

```
Login Succeeded
mustafakhattak@Mustafas-MacBook-Air-4 CS651Project1 % cd /Users/mustafakhattak/Documents/GitHub/CS651Project1
mustafakhattak@Mustafas-MacBook-Air-4 CS651Project1 % pwd
ls -la
/Users/mustafakhattak/Documents/GitHub/CS651Project1
total 160
drwxr-xr-x@ 15 mustafakhattak  staff    480 Mar  2 23:05 .
drwxr-xr-x@  4 mustafakhattak  staff    128 Feb 21 05:48 ..
-rw-r--r--@  1 mustafakhattak  staff     41 Mar  2 23:05 .dockerignore
-rw-r--r--@  1 mustafakhattak  staff   8196 Mar  2 21:49 .DS_Store
drwxr-xr-x@ 15 mustafakhattak  staff    480 Mar  2 22:59 .git
-rw-r--r--@  1 mustafakhattak  staff   8797 Mar  2 22:58 about.html
-rw-r--r--@  1 mustafakhattak  staff   8310 Mar  2 22:58 contact.html
drwxr-xr-x   3 mustafakhattak  staff     96 Feb 20 10:39 css
-rw-r--r--@  1 mustafakhattak  staff   6669 Mar  2 22:59 dashboard.html
-rw-r--r--@  1 mustafakhattak  staff    224 Mar  2 23:05 Dockerfile
drwxr-xr-x  17 mustafakhattak  staff    544 Mar  2 22:56 Images
-rw-r--r--@  1 mustafakhattak  staff   9551 Mar  2 22:06 index.html
drwxr-xr-x   3 mustafakhattak  staff     96 Feb 20 10:40 js
drwxr-xr-x   2 mustafakhattak  staff     64 Feb 20 09:57 signin
-rw-r--r--@  1 mustafakhattak  staff  15475 Mar  2 22:59 signin.html
mustafakhattak@Mustafas-MacBook-Air-4 CS651Project1 %
```

Figure 6: Verification of the VisionForge project directory before building the Docker image. The output confirms the presence of the Dockerfile, .dockerignore, HTML pages, CSS, JavaScript files, and image assets required for containerization.

Screenshot #7: Docker Image Build from Dockerfile (Local Environment)

```
Login Succeeded
mustafakhattak@Mustafas-MacBook-Air-4 CS651Project1 % cd /Users/mustafakhattak/Documents/GitHub/CS651Project1
mustafakhattak@Mustafas-MacBook-Air-4 CS651Project1 % pwd
ls -la
/Users/mustafakhattak/Documents/GitHub/CS651Project1
total 160
drwxr-xr-x@ 15 mustafakhattak  staff    480 Mar  2 23:05 .
drwxr-xr-x@  4 mustafakhattak  staff    128 Feb 21 05:48 ..
-rw-r--r--@  1 mustafakhattak  staff     41 Mar  2 23:05 .dockerignore
-rw-r--r--@  1 mustafakhattak  staff   8196 Mar  2 21:49 .DS_Store
drwxr-xr-x@ 15 mustafakhattak  staff    480 Mar  2 22:59 .git
-rw-r--r--@  1 mustafakhattak  staff   8797 Mar  2 22:58 about.html
-rw-r--r--@  1 mustafakhattak  staff   8310 Mar  2 22:58 contact.html
drwxr-xr-x   3 mustafakhattak  staff     96 Feb 20 10:39 css
-rw-r--r--@  1 mustafakhattak  staff   6669 Mar  2 22:59 dashboard.html
-rw-r--r--@  1 mustafakhattak  staff    224 Mar  2 23:05 Dockerfile
drwxr-xr-x  17 mustafakhattak  staff    544 Mar  2 22:56 Images
-rw-r--r--@  1 mustafakhattak  staff   9551 Mar  2 22:06 index.html
drwxr-xr-x   3 mustafakhattak  staff     96 Feb 20 10:40 js
drwxr-xr-x   2 mustafakhattak  staff     64 Feb 20 09:57 signin
-rw-r--r--@  1 mustafakhattak  staff  15475 Mar  2 22:59 signin.html
mustafakhattak@Mustafas-MacBook-Air-4 CS651Project1 % docker build -t visionforge-site:latest .
[+] Building 1.1s (8/8) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 263B
 => [internal] load metadata for docker.io/library/nginx:alpine
 => [auth] library/nginx:pull token for registry-1.docker.io
 => [internal] load .dockerignore
 => => transferring context: 81B
 => [internal] load build context
 => => transferring context: 993B
 => [1/2] FROM docker.io/library/nginx:alpine@sha256:1d13701a5f9f3fb01aaa88cef2344d65b6b5bf6b7d9fa4cf0dca557a8d7702ba
 => CACHED [2/2] COPY . /usr/share/nginx/html
 => exporting to image
 => => exporting layers
 => => writing image sha256:9fbe8303f0fa839818876cbc503b908a0712a37c7765a0e0760d7e30e54d737f
 => => naming to docker.io/library/visionforge-site:latest
WARNING: current commit information was not captured by the build: failed to read current commit information with git rev-parse --is-inside-work-tree

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/ynaxh377ws3kkkfcnd08e4xxo

What's next:
    View a summary of image vulnerabilities and recommendations → docker scout quickview
mustafakhattak@Mustafas-MacBook-Air-4 CS651Project1 %
```

Figure 7: The VisionForge application was containerized using Docker. The image was successfully built using the nginx:alpine base image, and project files were copied into /usr/share/nginx/html. The output confirms successful image creation tagged as visionforg

Screenshot #8: Tagging and Pushing Docker Image to Docker Hub

```
[mustafakhattak@Mustafas-MacBook-Air-4 CS651Project1 % docker tag visionforge-site:latest mustafakhattak/visionforge-site:latest
[mustafakhattak@Mustafas-MacBook-Air-4 CS651Project1 % docker images | head -n 20
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
IMAGE                                          ID             DISK USAGE   CONTENT SIZE   EXTRA
docker/getting-started:latest                  289dc403af49       46.5MB             0B   U
hello-world:latest                             ca9905c726f0        5.2kB             0B   U
mcr.microsoft.com/mssql/server:2019-latest     361a3c9a0747       1.47GB             0B   U
moby/buildkit:buildx-stable-1                  a438d2bdd438        231MB             0B   U
mustafakhattak/f5-1-react:1.0                  69e90cc1aa94       62.3MB             0B
mustafakhattak/playground:latest               487fdd2f3060        290MB             0B   U
mustafakhattak/playground:v1.0                 b840f03f2e5b        289MB             0B
mustafakhattak/visionforge-site:latest         9fbe8303f0fa       70.3MB             0B   U
vf-sms-twilio:latest                           487fdd2f3060        290MB             0B   U
visionforge-site:latest                        9fbe8303f0fa       70.3MB             0B   U
[mustafakhattak@Mustafas-MacBook-Air-4 CS651Project1 % docker push mustafakhattak/visionforge-site:latest
The push refers to repository [docker.io/mustafakhattak/visionforge-site]
8da38635b2be: Pushed
5196ca0ce600: Mounted from library/nginx
a5aee8cb909c: Mounted from library/nginx
e7a74677e0bc: Mounted from library/nginx
ca1e0477244e: Mounted from library/nginx
b55a1191629e: Mounted from library/nginx
ce442f0e19c5: Mounted from library/nginx
cfbf14f957f1: Mounted from library/nginx
45f3ea5848e8: Mounted from library/nginx
latest: digest: sha256:14f7202a879c65b37385ddb48c0d504099e8cc82237fc2bc03bb8dc1c9441ff3 size: 2200
mustafakhattak@Mustafas-MacBook-Air-4 CS651Project1 %
```

Figure 8: The locally built Docker image was tagged with the Docker Hub repository name mustafakhattak/visionforge-site:latest and successfully pushed to Docker Hub. The output confirms layer uploads and digest generation, making the image publicly accessible fo

Screenshot #9: Docker Hub Repository Confirmation - visionforge-site

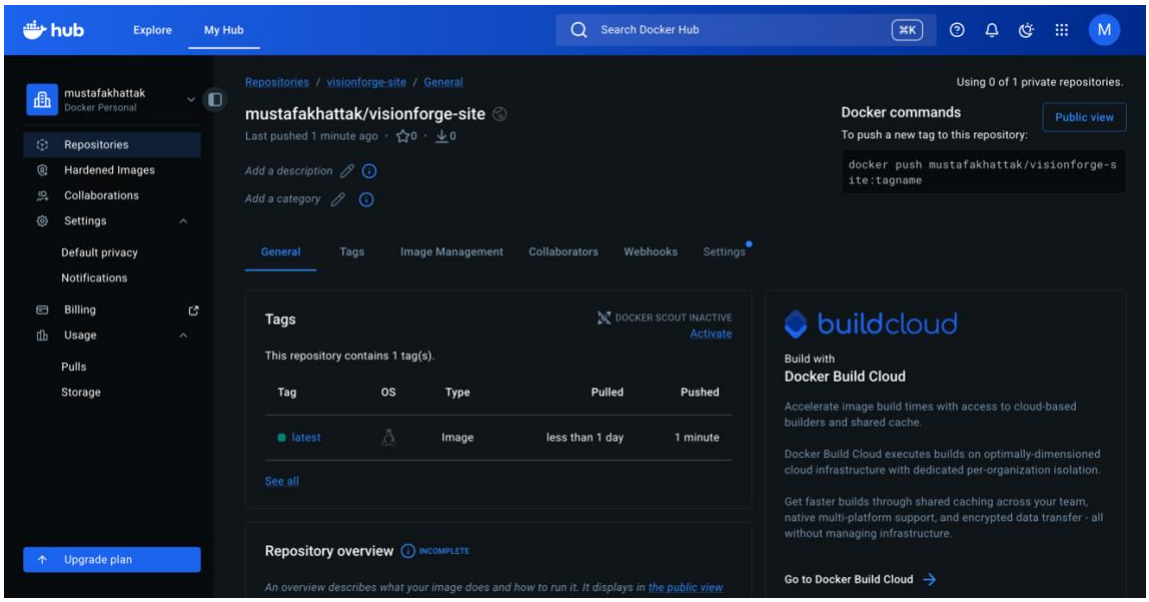

Figure 9: The Docker image mustafakhattak/visionforge-site:latest is successfully published to Docker Hub. The repository shows the latest tag, push timestamp, and image availability, confirming readiness for cloud deployment on AWS EC2.

Section 3 - AWS EC2 Deployment

In this section, the Dockerized VisionForge AI application is deployed to the cloud using Amazon Web Services (AWS) EC2. The goal is to move the application from a local development environment to a publicly accessible cloud server.

The deployment process includes launching an EC2 instance, configuring security groups, generating a key pair for secure SSH access, connecting to the instance, pulling the Docker image from Docker Hub, and running the container on port 80. This section demonstrates the complete cloud hosting workflow and verifies that the application is successfully accessible via a public IP address.

Screenshot #10: Launching AWS Academy Learner Lab Environment

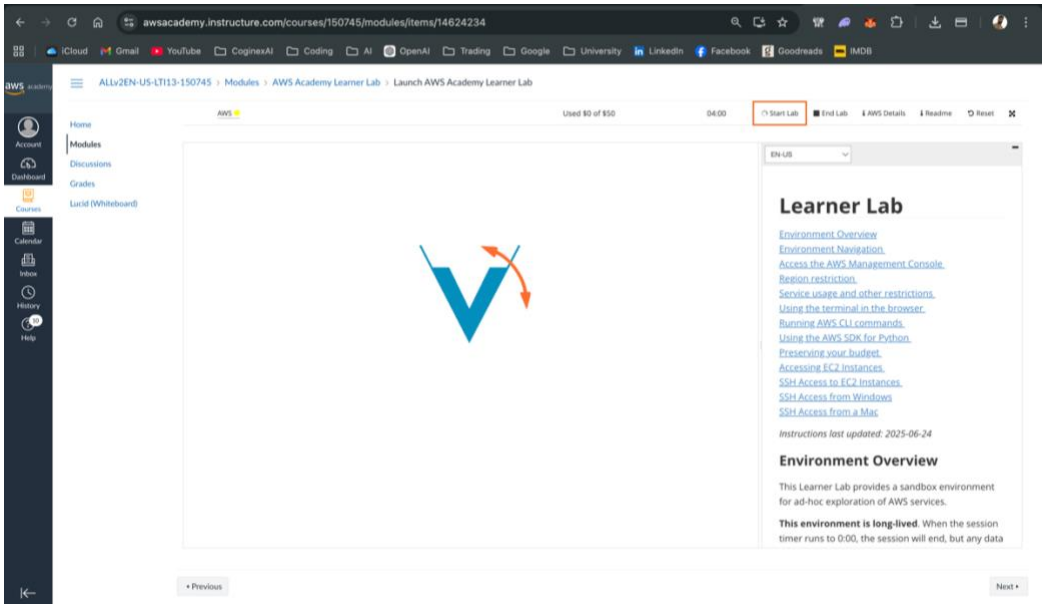

Figure 10: The AWS Academy Learner Lab environment was launched to provision a sandbox AWS account for deploying the VisionForge application. This environment provides access to EC2 and other AWS services required for cloud-based deployment.

Screenshot #11: AWS EC2 Dashboard - Preparing to Launch Instance

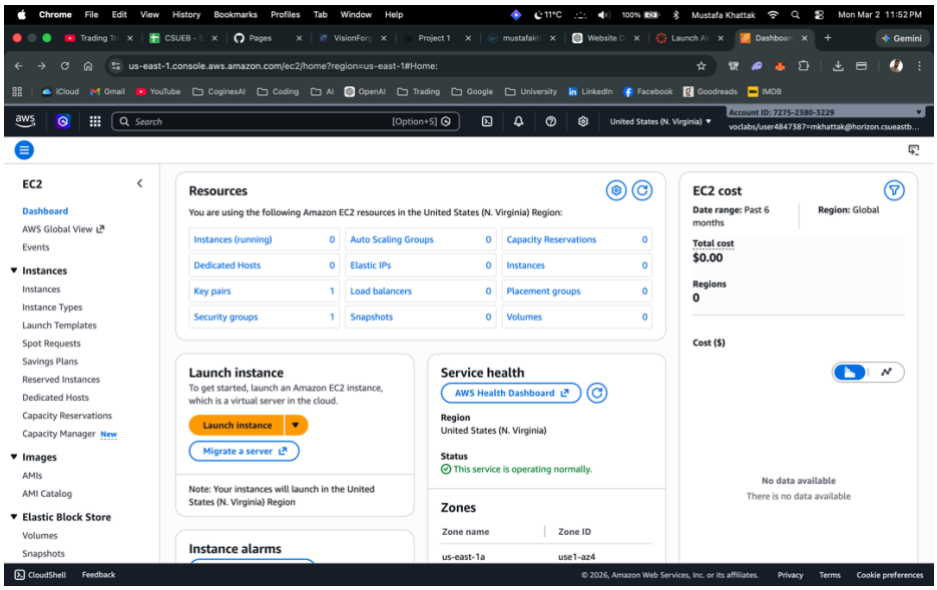

Figure 11: The AWS EC2 dashboard in the US East (N. Virginia) region is accessed to provision a new virtual server. The dashboard confirms no running instances and shows available resources prior to launching the EC2 instance for deployment.

Screenshot #12: Creating EC2 Key Pair for Secure SSH Access

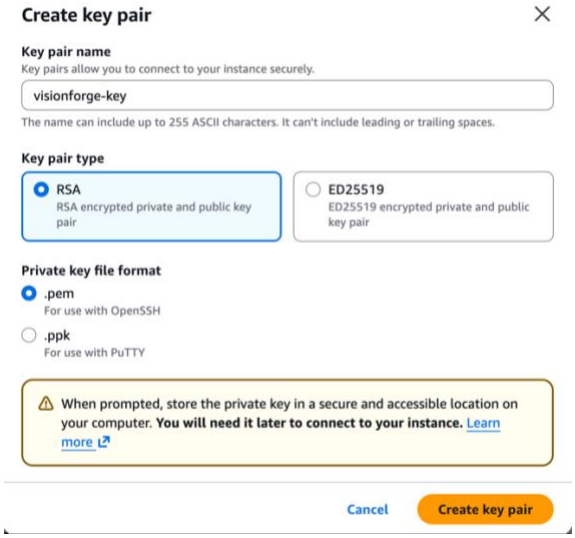

Figure 12: A new RSA key pair named visionforge-key was generated in .pem format to enable secure SSH access to the EC2 instance. The private key file is required to authenticate and connect to the virtual server from a local machine.

Screenshot #13: Configuring Security Group - Enabling SSH and HTTP Access

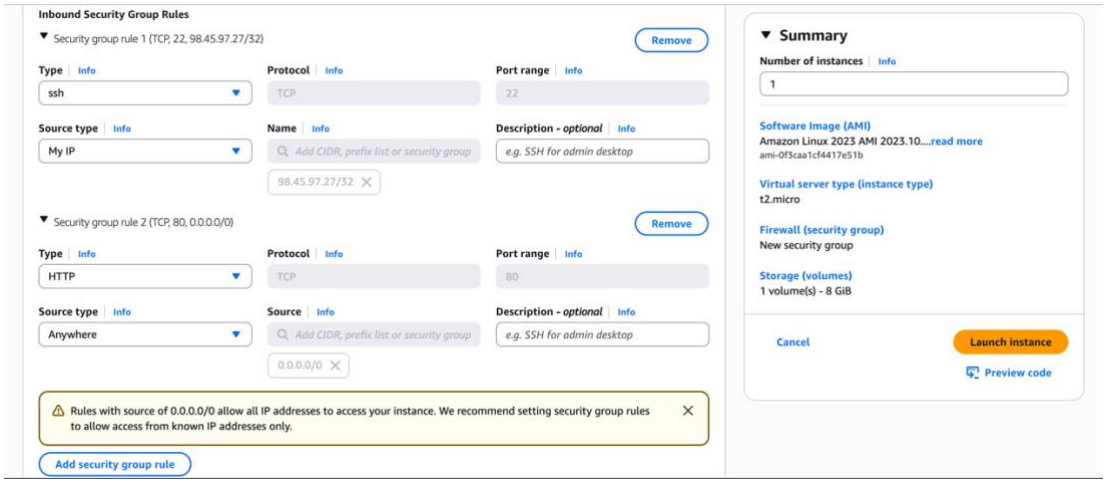

Figure 13: Inbound security group rules were configured to allow secure SSH access (port 22) from the developer's IP address and HTTP access (port 80) from anywhere (0.0.0.0/0). This configuration enables remote administration while allowing public access to the

Screenshot #14: EC2 Instance Successfully Launched - Running State Confirmation

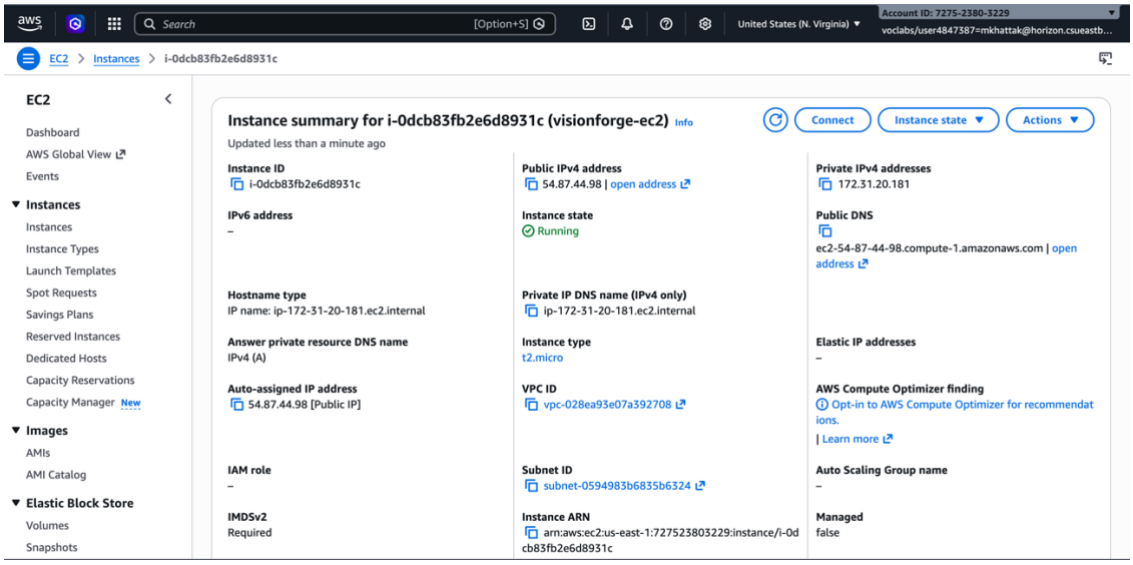

Figure 14: The EC2 instance visionforge-ec2 was successfully launched in the US East (N. Virginia) region. The instance is in a Running state and assigned a public IPv4 address (54.87.44.98), enabling remote SSH access and public web hosting.

Screenshot #15: Establishing Secure SSH Connection to EC2 Instance

```
Terminal  Shell  Edit  View  Window  Help

Last login: Mon Mar  2 23:21:01 on ttys002
  [Restored Mar 3, 2026 at 12:03:48 AM]
Last login: Mon Mar  2 23:38:37 on ttys000
Restored session: Mon Mar  2 23:38:40 PST 2026
mustafakhattak@Mustafas-MacBook-Air-4 ~ % cd ~/Downloads
mustafakhattak@Mustafas-MacBook-Air-4 Downloads % chmod 400 visionforge-key.pem
mustafakhattak@Mustafas-MacBook-Air-4 Downloads % ssh -i visionforge-key.pem ec2-user@54.87.44.98
The authenticity of host '54.87.44.98 (54.87.44.98)' can't be established.
ED25519 key fingerprint is: SHA256:My6LlkEWSXfHkMkr8C6YVDc406kqM1h+WXk3nsYvL28
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '54.87.44.98' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-20-181 ~]$
```

Figure 15: A secure SSH connection was established to the EC2 instance using the generated private key file (visionforge-key.pem). Successful login to the Amazon Linux 2023 environment confirms remote server access and readiness for Docker installation and applic

Screenshot #16: Pulling Docker Image from Docker Hub on EC2

```
[ec2-user@ip-172-31-20-181 ~]$ docker rm -f visionforge 2>/dev/null || true
[ec2-user@ip-172-31-20-181 ~]$ docker pull mustafakhattak/visionforge-site:latest
latest: Pulling from mustafakhattak/visionforge-site
589002ba0eae: Already exists
bca5d04786e1: Pull complete
3e2c181db1b0: Pull complete
6b7b6c7061b7: Pull complete
399d0898a94e: Pull complete
955a8478f9ac: Pull complete
6d397a54a185: Pull complete
5e7756927bef: Pull complete
0966b5e835a2: Pull complete
Digest: sha256:ba48bcd010a213179bb150cb2060f93476847ab4a19e47457d8617e8176a31c8
Status: Downloaded newer image for mustafakhattak/visionforge-site:latest
docker.io/mustafakhattak/visionforge-site:latest
[ec2-user@ip-172-31-20-181 ~]$
```

Figure 16: The Docker image mustafakhattak/visionforge-site:latest was successfully pulled from Docker Hub onto the EC2 instance. The output confirms all layers were downloaded and verified, preparing the container for execution on the cloud server.

Screenshot #17: Running VisionForge Container on EC2 and Verifying Active Status

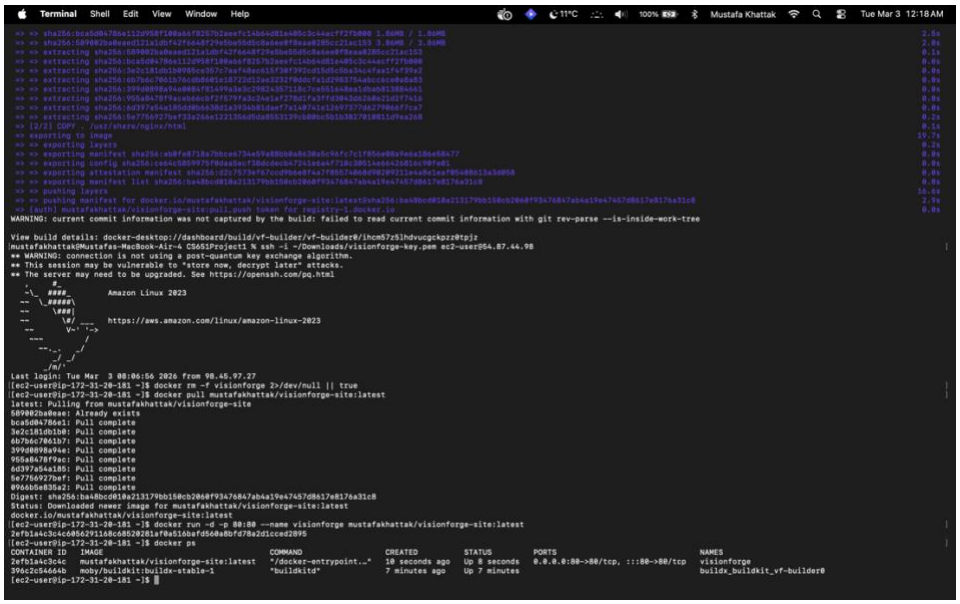

Figure 17: The VisionForge Docker container was launched on the EC2 instance using port mapping -p 80:80. The docker ps output confirms the container is running successfully and bound to port 80, making the web application accessible via the instance's public IP

Screenshot #18: VisionForge AI Successfully Deployed and Accessible via Public EC2 IP

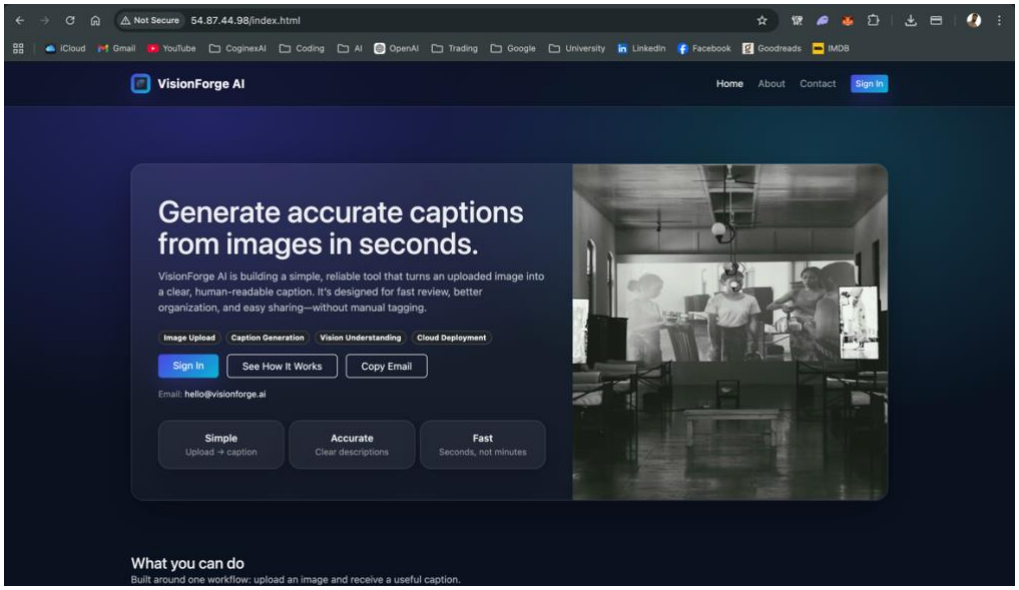

Figure 18: The VisionForge AI web application is successfully running on the AWS EC2 instance and is publicly accessible through the instance's IPv4 address (http://54.87.44.98. This confirms correct Docker container execution, security group configuration, and successful cloud deployment.

Final Deployment Summary

VisionForge AI was successfully containerized using Docker and deployed to an AWS EC2 instance. The Docker image was built locally, pushed to Docker Hub, and pulled onto the EC2 instance. After configuring the security group to allow HTTP traffic and running the container on port 80, the application became publicly accessible via the instance's public IP address.

This project demonstrates a complete end-to-end cloud deployment workflow including:

- Static web application development
- Docker containerization
- Docker Hub image publishing
- AWS EC2 provisioning
- Secure SSH access
- Security group configuration
- Production-style container deployment

The deployment was verified by accessing the application through a public browser using the EC2 public IPv4 address.